Mathematics and assumptions

Companion to the nine picture strips · constructions and their added premises

Reading the pictures

Read each strip left to right; row numbers here count from the top. Black and green distinguish contributions. Orange shows a result, transferred piece or rate as specified below. Grey shows a prior position or shift; arrows in the gutters indicate reading order. Drawn geometry is a chosen representation. No background axes, physical observer at infinity, or quantum amplitudes are required to read the pictures.

1–2 · Comparison, composition and cancellation

Supply an oriented affine plane and comparison by translation. Tip-to-tail joining defines vector addition; following a step by its reverse gives zero net displacement. WDP §§1.1–1.4 requires a declared comparison structure. Its §17.7 constructs signed histories after alternative formation and additive completion are admitted. Geometric reverse steps illustrate cancellation; they do not prove every physical operation reversible. WDP §17.8 constructs $g(a,b)=L(a)^{T}L(b)$ from a supplied additive report L, with radical ker L; that pairing is observer-relative, not selected by bare difference.

3 · A report can forget what the next move needs

Inputs $\psi\pm=(3,\pm4)/5$ have equal component-length reports (3/5,4/5). Apply the SAME orthogonal matrix $U=((3,4),(-4,3))/5$: $U\psi+=(1,0)$, $U\psi-=(-7,-24)/25$. The picture uses scale 1.5 throughout. Middle-panel grey is the old joined vector; orange is the rotated vector; the last panel resolves it into components. Equal old reports give different new ones, so no report-only update works for both. A supplied square report gives (9/25,16/25) initially and first outputs 1 versus 49/625. This is not Born uniqueness, nor a failure of the l1 norm to retain signs. Sources: FQM §13.8; Proofs §§1.2–1.4.

4–5 · Balance first; a quarter-turn as an added move

Redistribute lengths (3,1)→(2,2); both total 4. More generally $z_{k+1}=z_k+d_0^* j_k$ preserves the vertex sum because each internal edge contributes opposite signs. For a closed cycle γ, $\langle A+d_0\lambda,\gamma\rangle=\langle A,\gamma\rangle$. These are balance and repair-invariance identities, not Noether's theorem (WDP §5.12). Row 5 supplies $J(x,y)=(-y,x)$: $J^2=-I$ and $J^4=I$. With an admitted rational pairing g and $g(Jx,y)=-g(x,Jy)$, WDP §17.9 derives the Q(i) action $(a+ib)x=ax+bJx$ and Hermitian form $h=g+i\,g(J\cdot,\cdot)$. J and its compatibility are inputs. Repeated direction is not elapsed time; rational turns complete to a circle, not the time line.

6 · The extra structure for classical Hamilton

Now supply a STATE plane with dimensionless conjugate coordinates q,p and area $\omega=dq\wedge dp$. The two triangles form a patch of area 5.4; the shown shears preserve it. In rows 7–9 black is q, green is p; right/up are positive. Their right angle does not derive conjugacy. Assume a time-independent continuously differentiable flow preserves EVERY small oriented area: $\partial q\, u+\partial p\, v=0$. Thus $\alpha=u\,dp-v\,dq$ is closed; on a simply connected patch $\alpha=dH$, giving $\dot q=H_p$ and $\dot p=-Hq$. H is local, unique up to a constant. The sampled shears illustrate this hypothesis; they do not establish it for arbitrary flows.

7 · A continuous rule makes the arrow turn

Choose $H=(q^2+p^2)/2$, hence $\dot q=p$ and $\dot p=-q$ simultaneously. Both rate panels show (2.1,1.2); orange rate lengths are 1.2 and 2.1 on the same scale, not successive kicks. The last panel shows exact states at t=0,0.5,1: $q(t)=q_0\cos t+p_0\sin t$, $p(t)=p_0\cos t-q_0\sin t$. It preserves H and ω. Smaller snapshots alone do not prove continuity or select a generator. This clockwise oscillator uses −J from row 5; reversing the orientation is explicit.

8–9 · A different rule, its symmetry, and the Noether charge

Switch to $H=p^2/2$: $\Phi t(q,p)=(q+tp,p)$. Translations $Ta(q,p)=(q+a,p)$ preserve ω and H, and commute with Φt. Both orderings start at (2,0.65), use a=1.5, t=1, and end at (4.15,0.65); grey means shift, orange means slide. Row 9 is the unshifted run t=0,1,2. Translation pairs with ω to give dp, so its charge is p; $\dot p=-Hq=0$. For $S=\int(p\dot q-H)dt$, independent interior variations with $\delta q=0$ at endpoints give Hamilton's equations. Constant translations preserve S and map endpoint data to translated data. Noether conservation holds on stationary trajectories; equal marks alone do not prove it.

The separate algebraic Hamiltonian route and remaining limits

WDP §19A.6 starts instead with K=Q(i), matrix multiplication, an adjoint and a K-linear derivation δ satisfying Leibniz and the adjoint law. $S=\Sigma_j\delta(E_{j1})E_{1j}$ gives $\delta=[S,\cdot]$; after skew-adjoint adjustment, $H_0=iS$ gives $\delta=-i[H_0,\cdot]$, unique modulo rational scalar identity. This constructs an algebraic generator, not a clock or flow, and is distinct from row 6's classical theorem. Neither route selects physical energy from marks. The sheets do not derive the Born rule, Everett, Bell, a unique Hilbert geometry or physical laws. Novice comprehension remains untested.